

PART 1: Analysis of a small data-set.

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1 Data-set used in the analysis

The present work analyses the fictional data-set 'DATA_SET_REFERENCE_EVAL.csv', which presents 400 measurements of HDL and LDL in patients across two planets (Earth and Venus) from two populations (tall and small), as seen in Table 1.

The measurement units were not given, but based on the magnitude of the numbers, HDL and LDL are assumed to be in milligrams per deciliter (mg/dL). No further information was given about the dataset due to it being fictional; nonetheless, in a real one, the source must be trustworthy with traceable methods.

Table 1: Fictional Data-Set. First 6 entries

	HDL	LDL	Size	Planet
Patient1	222.0001	114.1767	Tall	Venus
Patient2	150.3521	161.5871	Tall	Earth
Patient3	188.2625	114.2765	Tall	Venus

1.1 Overview

- **Total Records:** 400 patients.
- **Locations:** Distributed nearly evenly across two planets (204 on Venus, 196 on Earth).
- **Demographics:** Exactly split by size (200 Tall, 200 Small).
- **Columns:**
 - **Patient:** Unique identifier (Patient1 to Patient400).
 - **HDL (High-Density Lipoprotein levels):** Numerical variable (mg/dL).
 - **LDL (Low-Density Lipoprotein levels) :** Numerical variable (mg/dL).
 - **Size:** Categorical variable (Tall or Small).
 - **Planet:** Categorical variable (Venus or Earth).

Both columns, HDL and LDL, contain negative values (something impossible) and one NA value each, telling us that the data must be cleaned. Apart from that, all the other columns seem to be in order, without typos or strange categories; therefore, most of the attention will be centered on the columns HDL and LDL.

2 Curation of the data

2.1 Negative and NA values

It was decided to remove each patient who has a negative or NA value. For this particular data-set, both negative and NA values for HDL and LDL corresponded to only 2 patients as seen in Table 2.

Table 2: Negative and NA Values

	HDL	LDL	Size	Planet
Patient8	NA	-30	Tall	Earth
Patient11	-20	NA	Tall	Earth

2.2 Outliers

Outliers were identified using a threshold of three standard deviations from the mean. For each numerical column in the dataset, we calculated the upper limit (Equation 1) and lower limit (Equation 2) as follows:

$$\text{Upper Limit} = \text{mean} + 3 \times \text{SD} \quad (1)$$

$$\text{Lower Limit} = \text{mean} - 3 \times \text{SD} \quad (2)$$

However, after applying these filters, no values were excluded, resulting in the same 398 observations remaining from the 'Negative and NA values' step.

3 Data exploration

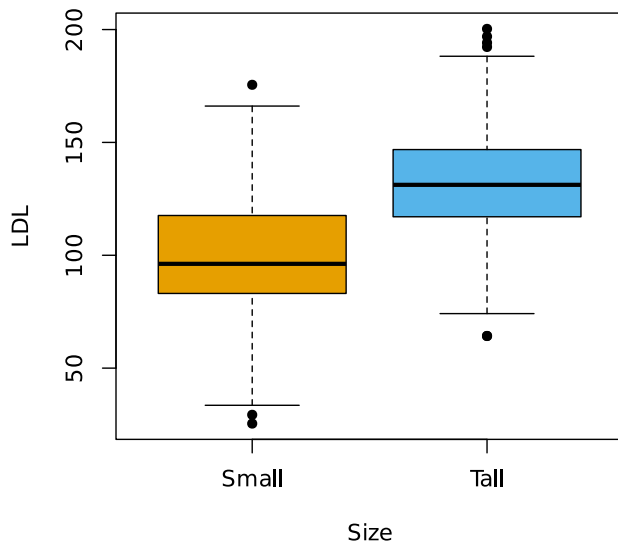


Figure 1: LDL levels by Size group

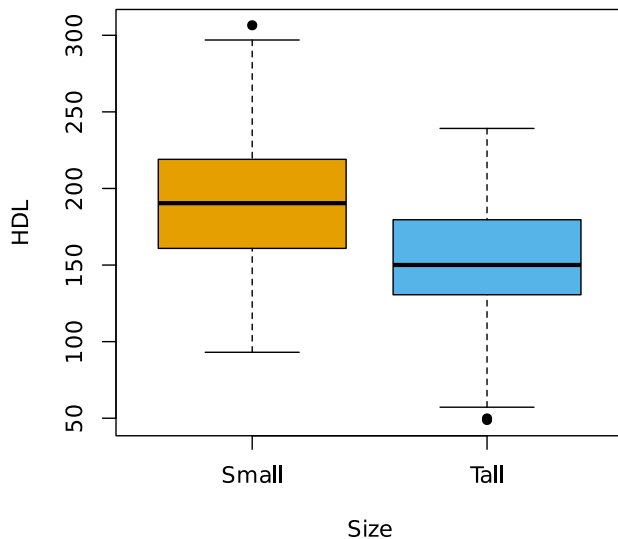


Figure 2: B. HDL levels by Size group

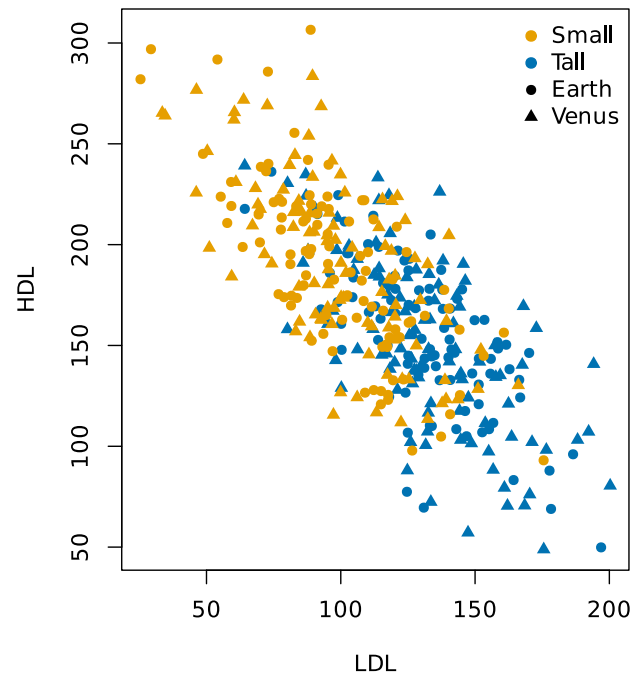


Figure 3: HDL vs LDL by Size (color) and Planet (shape)

4 Statistics used to validate associations.

4.1 (a) 1 analysis of 2 continuous variables

Table 3: Shapiro-Wilk normality test for HDL and LDL

Variable	Statistic	p-value	Conclusion
HDL	0.997	7.25e-01	Fail to reject H0 (normal)
LDL	0.997	8.09e-01	Fail to reject H0 (normal)

Table 4: Correlation between HDL and LDL

Method	Correlation coefficient	p-value
Pearson	-0.745	1.24e-71

4.2 (b) 1 analysis of 1 continuous variable vs 1 categorical variable.

Table 5: Comparison groups

Comparison	Test	Statistic	p-value	Conclusion
Tall vs Small (HDL)	T-test	-9.053	6.46e-18	Sig.
Tall vs Small (LDL)	T-test	12.899	5.12e-32	Sig.
Earth vs Venus (HDL)	T-test	0.212	8.32e-01	N.S.
Earth vs Venus (LDL)	T-test	-0.270	7.87e-01	N.S.